

Bayesian Estimation of Causal Effects Using Proxies of a Latent Interference Network

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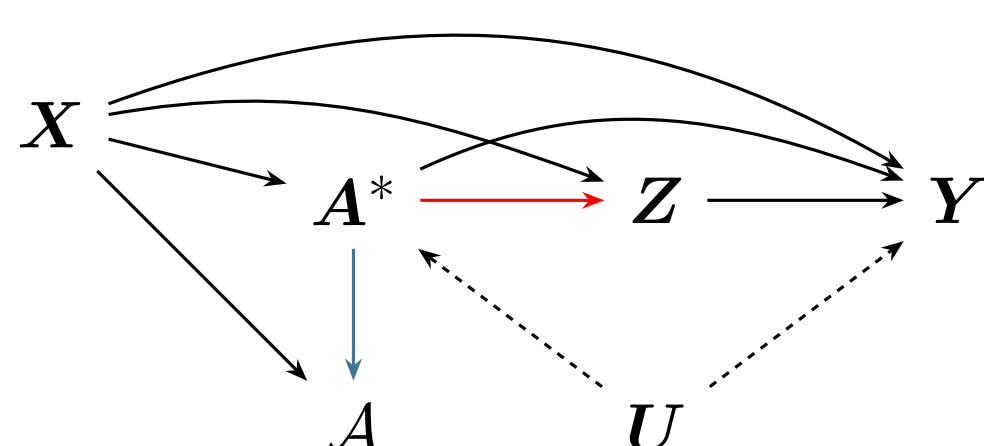
Introduction

- **Interference** = Outcome of a unit depends on treatments of others.
- Interference structure represented by a network.
- Accurately measuring networks of social interactions is a formidable task.
- Only proxy measurements are typically available.
 - Measurement error.
 - Networks from multiple sources.
 - Multilayer networks.
- **Example:** Evaluating vaccine or marketing spillovers where physical contacts are latent, but online friendships are observed.

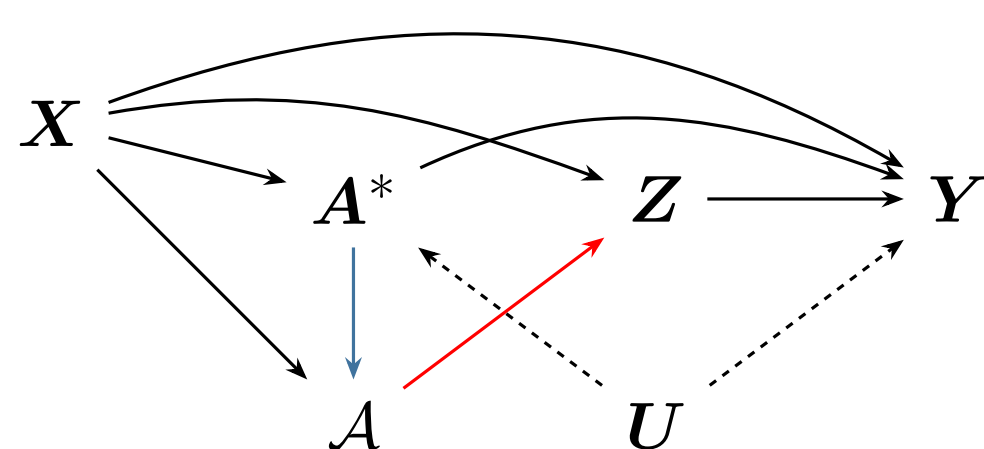
Setup and Structural Causal Models

- Finite population of N units. $\mathbf{Z}$ treatments; $\mathbf{Y}$ outcomes; $\mathbf{X}$ covariates.
- True (latent) interference network $\mathbf{A}^*$.
- Observed proxy networks $\mathcal{A} = \{\mathbf{A}_1, \dots, \mathbf{A}_B\}$.
- Observed data $\mathcal{D} = (\mathbf{Y}, \mathbf{Z}, \mathcal{A}, \mathbf{X})$.
- Outcome model

$$Y_i = f_Y(Z_i, \mathbf{X}_i, \mathbf{Z}_{-i}, \mathbf{X}_{-i}, \mathbf{A}^*, U_i, \varepsilon_{Y_i}).$$



(a) Causal proxies & Latent assignments



(b) Causal proxies & Proxy assignment

Fig.1 DAGs representing the assumed SCMs with causal proxies and treatment assignment based on the latent network $\mathbf{A}^*$ or the proxies $\mathcal{A}$. In the paper, we also consider non-causal proxies where $\mathcal{A}$ and $\mathbf{A}^*$ are both affected by a common cause $\mathbf{V}$.

Causal Estimands

- Effects of hypothetical interventions on population-level treatments.
- $Y_i(\mathbf{z})$ outcomes under the intervention $do(\mathbf{Z} = \mathbf{z})$.
- Treatment assignment policies:
 1. **Static policy.** Set $\mathbf{Z}$ to some fixed value $\mathbf{z}$.
 2. **Dynamic policy.** Set $\mathbf{Z}$ as a function of covariates $\mathbf{z} = h(\mathbf{X})$.
 3. **Stochastic policy.** $do(\mathbf{Z} = \mathbf{z})$ occurs with probability $\pi_a(\mathbf{z})$.
- Estimands are contrasts of

$$N^{-1} \sum_{i=1}^N \mathbb{E}[Y_i(\mathbf{z}) | \mathbf{A}^*, \mathbf{X}]$$
 for different policies.

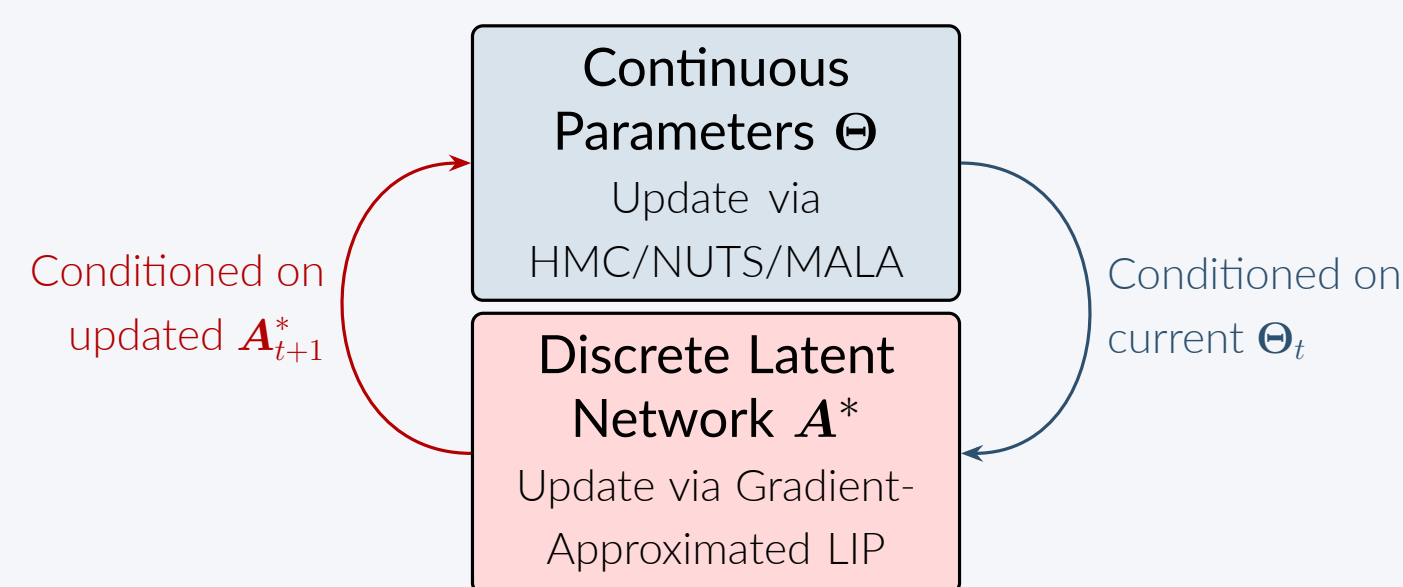
Identifiability of Structural Parameters

- **Global identifiability** in simplified settings.
 - Base case with one proxy $\Rightarrow$ **non identifiable**.
 - **Identifiability with:** at least two proxies; or network-dependent treatments; or network-correlated outcomes.
 - **Insight:** utilizing information from all data modules is crucial.
- Sufficient conditions for **local identifiability**.

Bayesian Estimation and Inference

Joint posterior distribution: $p(\Theta, \mathbf{A}^* | \mathcal{D})$.

- **The Challenge:** Mixed space of continuous parameters Θ and a massive discrete latent variables $\mathbf{A}^*$ containing $\mathcal{O}(2^{N^2})$ terms.
- **Why Standard Methods Fail:**
 - Marginalization over $\mathbf{A}^*$ is not possible.
 - Continuous relaxations performed poorly.
 - Random-walk MH or other MCMC scale badly.
- **Our Approach:** An end-to-end Block Gibbs MCMC algorithm that alternates between continuous Θ and discrete $\mathbf{A}^*$ updates.



Locally Informed Proposals

- Update $\mathbf{A}^*$ by flipping edges $A_{ij}^* = 0/1 \rightarrow A_{ij}^* = 1/0$.
- Flipping at random is inefficient.
- **Solution (LIP).** Proposal distribution that biases flips towards states with higher posterior probability:

$$\text{Proposal} \propto \exp\left(\frac{1}{2}\Delta(\mathbf{A}_{t+1}^*, \mathbf{A}_t^* | \cdot)\right) \mathbb{I}(\mathbf{A}_{t+1}^* \in H(\mathbf{A}_t^*)),$$
 where $\Delta(\mathbf{A}_{t+1}^*, \mathbf{A}_t^* | \cdot) = \log p(\mathbf{A}_{t+1}^* | \cdot) - \log p(\mathbf{A}_t^* | \cdot)$ and $H(\cdot)$ is the Hamming ball of size 1.
- Uses information from all modules to reconstruct the latent network $\mathbf{A}^*$.
- Exact LIP entails $\mathcal{O}(N^2)$ posterior evaluations per iteration $\Rightarrow$ **computationally prohibitive**.
- **Approximation** of posterior differences with one gradient evaluation per iteration:

$$\tilde{\Delta}(\mathbf{A}_t^*) = -(2\text{vec}(\mathbf{A}_t^*) - 1) \odot \nabla \log p(\mathbf{A}_t^*) \approx \Delta(\mathbf{A}_{t+1}^*, \mathbf{A}_t^*).$$
- **Formal analysis of approximation error.** Inversely related to interference signal-to-noise ratio but independent of N .
- Robust initialization strategy grounded in Bayesian modularization.

Numerical Illustration

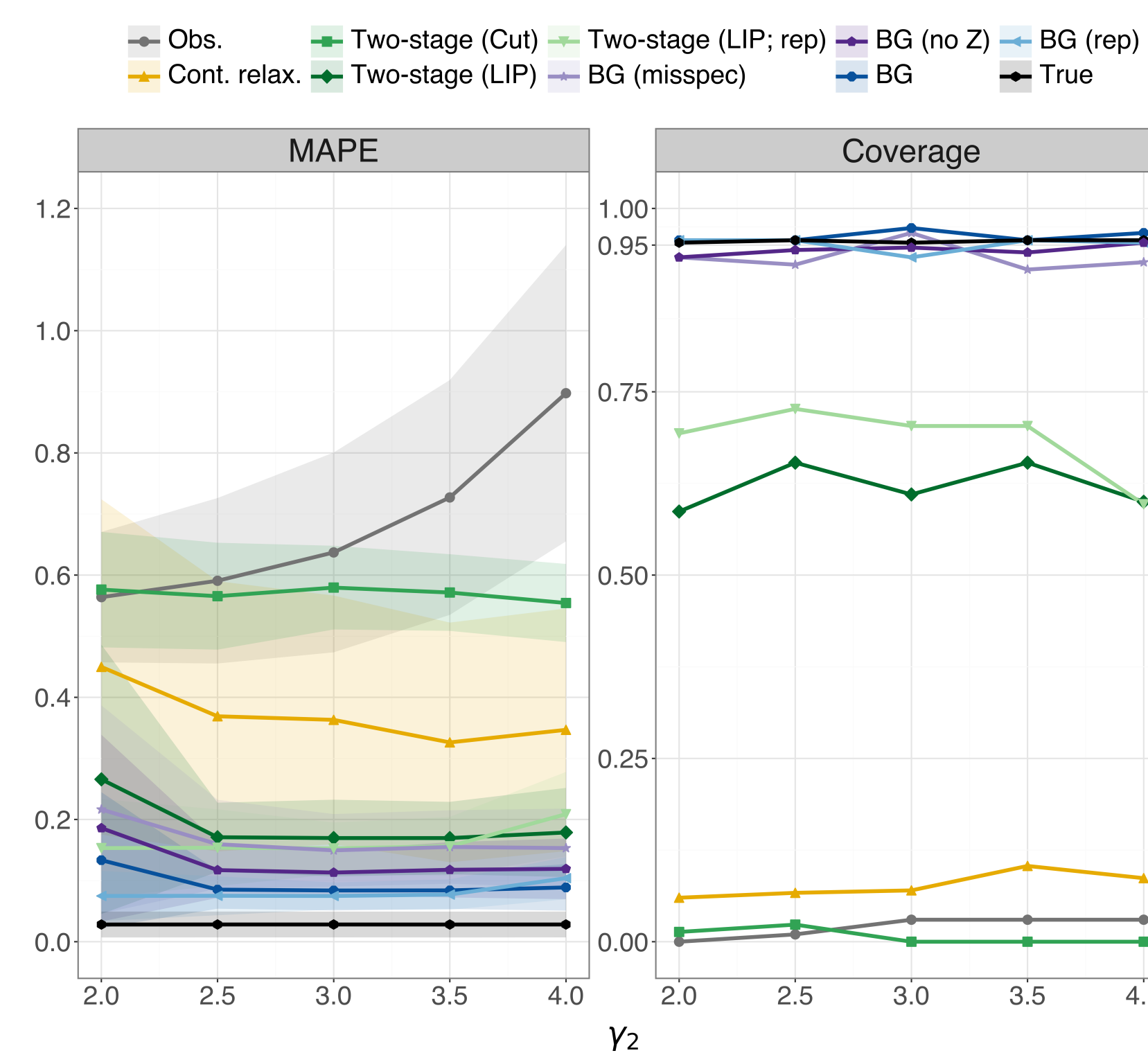


Fig.2 Mean Absolute Percentage Error and empirical coverage in estimating Total Treatment Effect (treating all vs. none). Proxies $\mathcal{A}$ are weaker as γ_2 (x-axis) increases. BG refers to our proposed Block Gibbs sampler. Two-stage refers to a baseline that first estimates $\mathbf{A}^*$ and then plugs it in to the outcome model as a fixed variable.

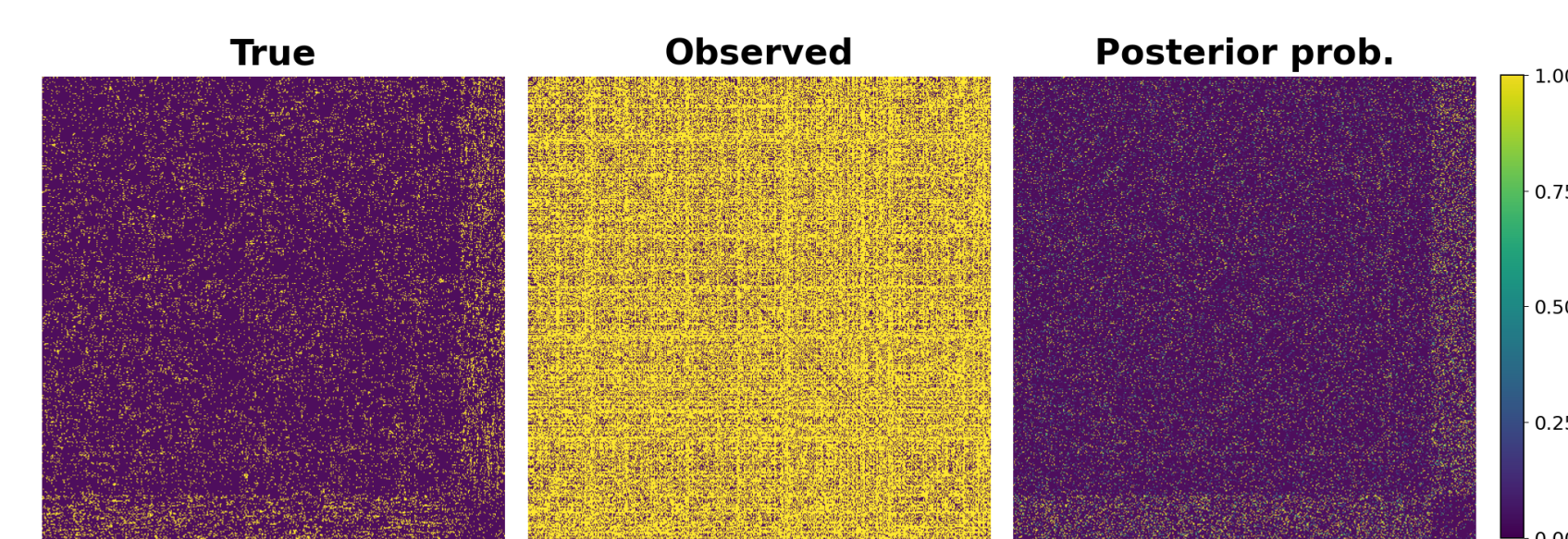


Fig.3 True network, observed proxy, and posterior probabilities in one iteration.

Key Contributions

- **Flexible structural causal models:** Unifies causal and non-causal proxy networks under diverse treatment assignment mechanisms.
- **End-to-end joint estimation:** Leverages information from all available data modules to simultaneously infer the latent network and causal effects.
- **Uncertainty propagation:** Accounts for and propagates uncertainty from the network inference stage to the final causal estimates.
- **Scalable discrete inference:** Employs gradient-approximated LIP to bypass the $\mathcal{O}(N^2)$ bottleneck and efficiently explore the massive discrete spaces.

